

Executive Summary



Paxos Trust Company: NY-regulated financial institution

Trust Charter granted 2015, qualified custodian under New York Banking Law

Source: PDF p1; paxos.com; dfs.ny.gov



Integration via FIX protocol

Your brokerage connects via FIX 4.2 to enable Bitcoin (XBTUSD) and Ethereum (ETHUSD) trading

Source: PDF p1-3; developer.paxos.com/fix-api



Trading settlement: OFF-CHAIN

Instant (~85ms), irreversible settlement on Paxos internal ledger

Source: PDF p1, p5-7, p14



Withdrawals: ON-CHAIN

BTC: ~30 minutes (3 confirmations); ETH: ~3 minutes (12 confirmations)

Source: PDF p1, p9-10, p12-13



Compliance: SHARED responsibility

Your firm cannot rely solely on Paxos; independent controls required

Source: PDF p1-2, p13-14

Company Overview & Regulatory Status

Founded in 2012, Qualified Custodian Status

Paxos operates as a qualified custodian under New York Banking Law

 Source: PDF p1; paxos.com/regulatory-overview

NYDFS Trust Charter

Initially granted in 2015, renewed annually with ongoing oversight

 Source: PDF p1; dfs.ny.gov

Global Regulatory Oversight

NYDFS (New York), MAS (Singapore), FSRA (Abu Dhabi); OCC National Trust Charter application filed August 2025 (pending)

 Source: PDF p1; occ.treas.gov

Recent NYDFS Settlement (August 2025)

\$48.5M penalty (\$26.5M fine + \$22M compliance investment) for AML/BSA compliance deficiencies and KYC failures

 Source: PDF p2; dfs.ny.gov/press-release/dfs-settles-paxos

Implication for External Clients

You CANNOT rely solely on Paxos for compliance. You must implement independent monitoring and controls.

 Source: PDF p2

Integration Architecture (FIX, REST, WebSocket, Webhooks)

FIX 4.2 Protocol Connection

Binary TCP/IP connection; TargetCompID=ITBIT; Heartbeat=30s

 Source: PDF p2; developer.paxos.com/fix-api

Connection Endpoints

Sandbox: fix-sandbox.itbit.com:4101; Production: fix.itbit.com:4101

 Source: PDF p2

Session Management

Persist/increment sequence numbers; weekly reset; recover on connection drop

 Source: PDF p3

Supported Symbols

XBTUSD (Bitcoin/USD), ETHUSD (Ethereum/USD)

 Source: PDF p3; itbit.com

Order Parameters

Tick sizes: BTC 0.25, ETH 0.05; Max market order: \$500k; Max limit: \$1.5M; Post-only available

 Source: PDF p3; developer.paxos.com

Complementary APIs

REST for custody operations; WebSocket for market/account data; Webhooks for event notifications

 Source: PDF p15-16; developer.paxos.com

Trading Flow: Off-Chain Settlement (BTC Buy Example)



⚡ Key Insight: All trading settlement happens OFF-CHAIN in Paxos's internal database. No blockchain transactions are involved in trading. Settlement is instant (~85ms) and irreversible.

Sources: PDF p3-8, p14; <https://www.paxos.com/trading-terms-and-conditions>

Note: Total end-to-end latency target ~85ms (PDF p14)

Trade Execution Deep Dive



Price-Time Priority Matching

Orders matched based on price first, then time of arrival; multiple fills possible for large orders

 Source: PDF p5



Multi-Fill Execution Example

Example: 1.0 BTC order filled as 0.3 BTC @ \$52,505.00 + 0.7 BTC @ \$52,506.00 = VWAP \$52,505.59

 Source: PDF p5



Atomic Ledger Settlement

Atomic database transaction updates buyer/seller balances, orders, and trade history simultaneously

Settlement is instant, off-chain, irreversible, with no blockchain reorg risk

 Source: PDF p6-7



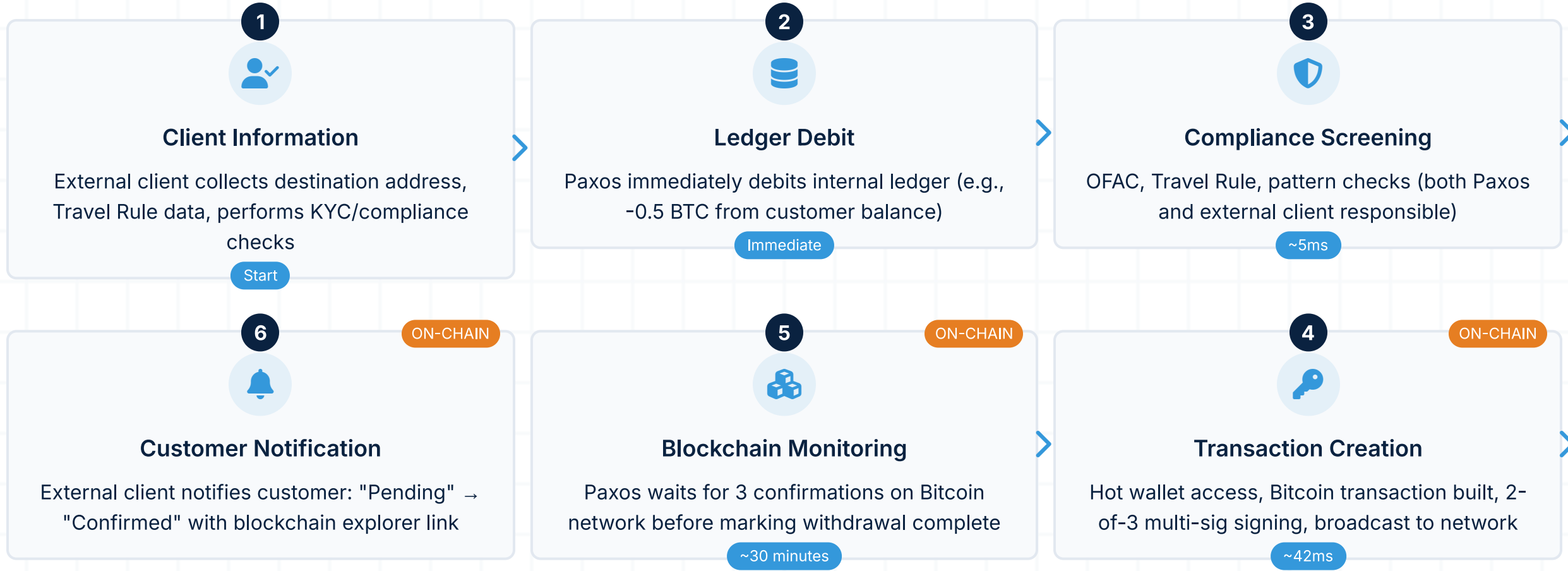
Execution Reports via FIX

External client receives FIX MsgType 8, ExecType partial/fill with prices, quantities, and status

Example: Partial Fill (0.3 BTC @ \$52,505.00), then Complete Fill (0.7 BTC @ \$52,506.00)

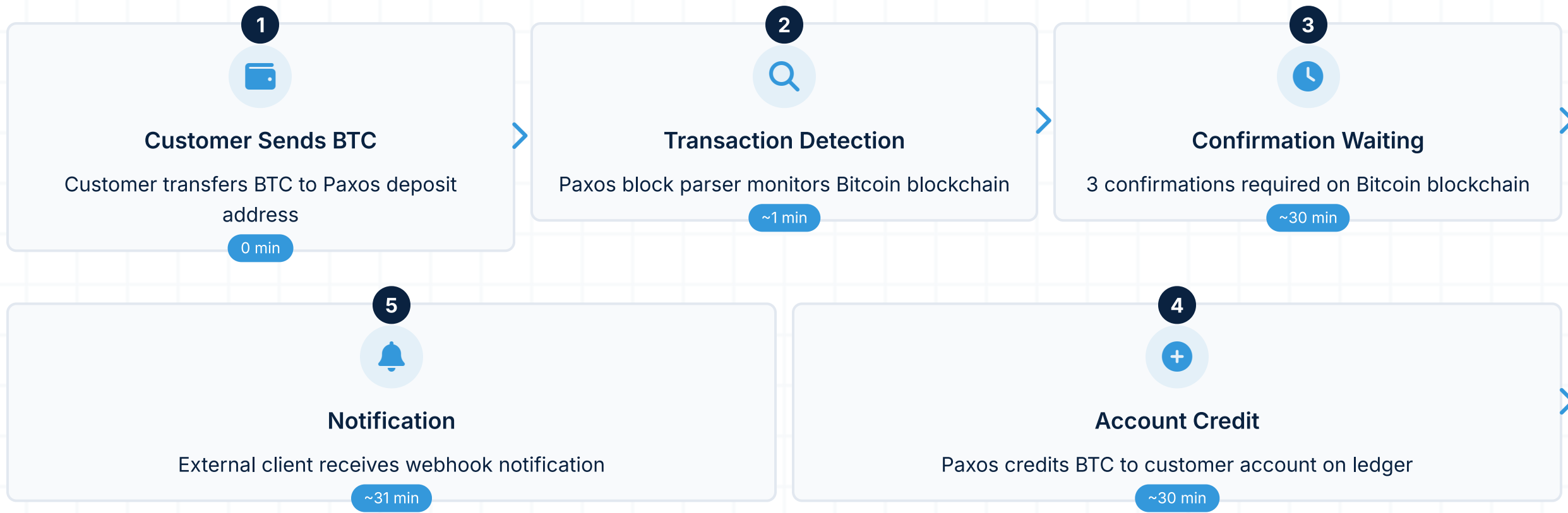
 Source: PDF p7

Withdrawal Process (Bitcoin, On-Chain)



⚠ Key Distinction:
Unlike trading (off-chain, ~85ms), withdrawals require blockchain confirmations. Internal processing takes ~100ms, but Bitcoin network finality requires 3 confirmations (~30 minutes).

Deposit Process (Bitcoin, On-Chain)



⚡ Key Insight:
Unlike trading (which is instantaneous), Bitcoin deposits require on-chain confirmation. Customers must wait for 3 Bitcoin confirmations (~30 minutes total) before the funds are credited to their account and available for trading.

Custody & Storage Architecture

Hot Wallet (~5% of total assets)

AWS-hosted, HSM-backed keys, 2-of-3 multi-signature, auto-rebalanced for fast withdrawals

 Source: PDF p11; paxos.com/security

Cold Storage (~95% of total assets)

Air-gapped servers in Faraday-cage vaults, 2-of-5 multi-signature, manual human approval required

 Source: PDF p11; paxos.com/security

Physical Security Specifications

DoD-certified facilities, military-grade encryption, geographic distribution of keys

 Source: PDF p11; paxos.com/security

Omnibus Wallet Structure

Customer assets exist on blockchain in Paxos omnibus wallets; individual entitlements tracked via internal ledger database

 Source: PDF p11-12

Custody Model

Paxos holds all private keys as qualified custodian; external clients do not hold keys; customers own proportional share of omnibus wallet

 Source: PDF p1, p12

Bitcoin vs Ethereum: Trading and Withdrawal

Bitcoin (XBTUSD)

 **Trading (Off-Chain)**
Instant settlement (~85ms)
Price-time priority matching

 **Withdrawal (On-Chain)**
3 confirmations required
~30 minutes to finality

 **Address Format**
bc1q... format (native SegWit)

 **Network Fees**
Variable (2-100+ satoshi/byte)
Based on network congestion

Ethereum (ETHUSD)

 **Trading (Off-Chain)**
Instant settlement (~85ms)
Price-time priority matching

 **Withdrawal (On-Chain)**
12 confirmations required
~3 minutes to finality

 **Address Format**
0x... format

 **Network Fees**
Variable (20-200 gwei)
Based on network congestion

Client Communication Guidance:

External clients should inform customers: "Bitcoin withdrawals take ~30 minutes; Ethereum withdrawals take ~3 minutes. Both times are blockchain confirmation times, not Paxos processing."

Compliance Responsibilities: You vs Paxos



What Paxos Handles:

- OFAC sanctions screening for transactions
- Suspicious activity detection on platform
- SAR filing (if detected by Paxos)
- Real-time transaction monitoring
- KYC services (available via API)

📄 Source: PDF p13-14; paxos.com/aml-kyc-disclosure



What External Clients MUST Do:

- Maintain your own AML/BSA program
- Implement CIP (Customer Identification Program)
- Perform KYC due diligence
- Independent suspicious activity monitoring
- Travel Rule compliance
- 6-year recordkeeping (SEC Rule 17a-4)

📄 Source: PDF p13-14; sec.gov/rules; fincen.gov/travel-rule



NYDFS Settlement Context (\$48.5M, August 2025)

Paxos was fined for compliance failures including inadequate partner due diligence, deficient AML/BSA controls, and KYC failures. This settlement demonstrates that **relying on third parties does NOT eliminate YOUR obligations**.

📄 Source: PDF p2, p14, p18; dfs.ny.gov/press-release/dfs-settles-paxos



Regulatory Expectations

Regulators expect **dual oversight**: Your compliance + Paxos compliance. The settlement demonstrates each party must maintain independent controls. You will be held responsible by SEC and FinCEN for customer activity, regardless of Paxos's controls.

📄 Source: PDF p14; sec.gov/faq/crypto-assets

❗ **CRITICAL:** The recent NYDFS Settlement shows Paxos had compliance gaps. You CANNOT rely solely on Paxos for compliance obligations.

Latency Breakdown

⚡ Trade E2E (BTC/ETH buy): ~85ms total

From FIX order to execution report

Component breakdown:

FIX message transmission	5ms
FIX parsing	5ms
Authentication	10ms
Compliance screening	20ms
Order validation	5ms
Order book matching	10ms
Ledger atomic update	15ms
Execution report encoding	5ms
Response transmission	5ms

📄 Source: PDF p14

🔗 Withdrawal (BTC): ~100ms internal + blockchain

Internal processing (~100ms) + ~30 min blockchain (3 confirmations)

Internal components:

Ledger debit	1ms
Compliance check	5ms
Bitcoin transaction creation	2ms
Multi-sig HSM signing	30ms
Bitcoin broadcast	10ms
Transaction mempool entry	52ms

📄 Source: PDF p15

💬 Customer Communication Guidance

For external clients to explain to their customers:

- ✓ "BTC/ETH purchases settle instantly (less than 100ms)"
- 🕒 "Bitcoin withdrawals will be finalized in about 30 minutes"
- 🕒 "Ethereum withdrawals will be finalized in about 3 minutes"

📄 Source: PDF p14-15

System Monitoring: Paxos and You

What Paxos Monitors 24/7



Order Performance

Pending orders, average latency (P99 target <100ms), daily filled orders, rejection rate

Source: PDF p16; paxos.com/infrastructure



Blockchain Status

Block times, transaction/gas fees, pending withdrawals for both BTC and ETH networks

Source: PDF p16; paxos.com/infrastructure



Ledger Consistency

Customer BTC/ETH vs. on-chain BTC/ETH, reconciliation every 15 seconds, critical alerts for mismatches

Source: PDF p16-17; paxos.com/infrastructure



Compliance & System Health

SAR reviews, blocked transactions, high-risk flags, CPU/memory/disk, database replication, API errors

Source: PDF p17; paxos.com/infrastructure

What External Clients Should Monitor



Transaction Patterns

Customer trading volumes, unusual patterns, geographic anomalies, high-risk connections

Source: PDF p17; sec.gov/faq/crypto-assets



Compliance

KYC status, SARs, blocked transactions, Travel Rule compliance for withdrawals

Source: PDF p17; sec.gov/faq/crypto-assets



Operations

FIX connection status, failed orders/executions, withdrawal status, pending deposits

Source: PDF p17; sec.gov/faq/crypto-assets